

Name:

ID:

**Network and Security
Assessment I**

I. Questions

1. What is the difference between IP address and Mac address?
2. Among all network topologies, which topology that causes much more expense when we use the same amount of the nodes? Why?
3. Among 7 layers of OSI model, which layer that provide the security for network communication? Describe its function
4. What is the difference between stream cipher and block cipher Cryptography?
5. What is the difference between symmetric and asymmetric key Cryptography System?

II. Exercises

1. The output equals to the input +2:
 - i. 'yjc v ku aqwt pcog?'➔ What does it mean? And what will be your answer?
2. The output of the system equal to the input XOR with the key respectively.
 - i. If the input is [45, 65, 70, 32, 65] and the key is [65, 32, 70, 65, 45].➔ What will be the output? Describe the details process of its calculation
3. If the input of substitution box 48 bits:

00111110000111111100000001111111101110001100011

And the fix matrix:

[5, 6, 12, 13, 14, 0, 6, 2, 15, 15, 7, 9, 14, 12, 7, 3]
[2, 3, 11, 11, 11, 3, 1, 1, 7, 9, 3, 15, 15, 14, 8, 13]
[9, 0, 13, 11, 7, 15, 15, 13, 13, 9, 13, 12, 7, 14, 5, 8]
[4, 12, 4, 12, 3, 6, 13, 2, 14, 4, 6, 8, 14, 11, 9, 8]

4. There are a new startup company and it consists of 5 departments. Each department is needed to design with required host respectively: 200, 15, 32, 64 and 10 hosts.
➔ Make IP calculation to assign in this new company